Kristin Branson

Senior Group Leader

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Summary

I develop new and impactful ways to use computer vision (CV) and machine learning (ML) to gain insight into scientific questions. I do this by 1) finding new scientific questions that can be framed as CV&ML problems, 2) engineering practical and integrative solutions, and 3) making these systems usable by others so that these methods are broadly adopted and applied to new problems. I was a pioneer of video-based analysis of animal behavior, and my work helped establish this as an important and ubiquitous technique in neuroscience and biology more generally.

Education

University of California, San Diego — *La Jolla, CA*
Ph.D., M.S., Computer Science — *2007,2002*
Dissertation Title: *Tracking Multiple Mice through Severe Occlusions*
Advisers: Serge Belongie and Sanjoy Dasgupta

Harvard University — *Cambridge, MA*
A.B. *Cum Laude*, Computer Science — *2000*

Experience

HHMI Janelia Research Campus — *Ashburn, VA*
Senior Group Leader — *2017-present*
Head of Computation & Theory — *2017-2022*
Group Leader — *2010-2017*

California Institute of Technology — *Pasadena, CA*
Postdoctoral Researcher — *2007-2009*
Advisers: Pietro Perona and Michael Dickinson

NASA Intern
Graduate Student Researcher — *NASA Ames RC*
Research Institute for Advanced Computer Science — *Summer 2000 & 2001*
Undergraduate Researcher — *NASA Armstrong Flight RC*
NASA Academy Program — *Summer 1999*

Open-Source Software

I lead, co-developed, and co-maintain the following open-source pieces of software:

- APT: The Animal Part Tracker, `http://kristinbranson.github.io/APT/`
- BABAM: The Browsable Atlas of Behavior-Anatomy Maps, `https://kristinbranson.github.io/BABAM/`
- BIAS: Basic Image Acquisition Software `http://stuff.iorodeo.com/notes/bias/`
- JAABA: Janelia Automatic Animal Behavior Annotator, `http://jaaba.sourceforge.net/`
- Ctrax: The Caltech Multiple Walking Fly Tracker, `http://ctrax.sourceforge.net/`

Conferences Organized

- Multi-Agent Behavior Workshop, CVPR, June, 2022.
- 4D Cellular Physiology Reimagined: Theory as a Principal Component, September, 2021.
- Janelia Conference: Women in Computational Biology, November 2019.
- Bioimage Computing Workshop, CVPR, June 2019.
- Society for Neuroscience Virtual Conference on Machine Learning, June 2019.
- Janelia Junior Scientist Workshop on Machine Learning and Computer Vision, April 2019, October 2017, October 2016, October 2015.
- Janelia Conference: What Can Machine Learning Do for Neuroscience and Vice-Versa?, November 2010.

Honors and Awards

- Selected as one of Cell's "40 under 40" scientists in commemoration of their 40th anniversary, 2015.
- Faculty of 1000 Recommendation for JAABA, 2013.
- NASA Graduate Student Researcher Program Fellowship, 2003-2006.

Scientific Leadership

- Led a lab from 2010-present.
- Established the Computation and Theory Research Area at Janelia (10 labs) and served as inaugural Head from 2017-2022. Set the overall vision and direction for computational research at Janelia; recruited, made hiring and renewal decisions, and set budgets for Group Leaders; initiated development of Janelia's GPU cluster; initiated Janelia's Theory Fellow program.
- As part of Senior Leadership at Janelia, participated in institute-wide planning and decision-making, 2017-2022.
- Co-created the Janelia Diversity, Equity, and Inclusion Committee, 2018. Chaired committee 2018-2020. Participated 2020-2022. As part of this committee, we have researched and proposed initiatives to Janelia leadership, invited speakers to present at Janelia, and started a DEI reading group.
- Started, co-organized, and contributed to Computer Vision and Machine Learning weekly reading group and internal seminar series at Janelia, 2014-present.
- Started and co-organized the Janelia Computation and Theory Seminar series, bi-weekly, 2018-present.
- Participated in and led a Women and Non-binary Scientist Mentoring Group at Janelia, 2018-present.
- Started and co-organized Janelia Computation and Theory social event to build community at Janelia, 2018-2020.

Outreach

- Directed the Cajal Machine Learning for Neuroscience Summer School (Summer, 2023).
- Taught classes on quantitative analysis of fly behavior at the Marine Biology Lab (June-July, 2022) and Howard University, an HBCU (September-November, 2022).
- Started and organized "Hour of Code" bi-monthly outreach activity for Janelia scientists to share their enthusiasm for and the impact of programming with grade-school children at local elementary schools and libraries, 2017-2020. This was the first student outreach activity at Janelia, and instigated Janelia to build a Community Relations group to support future efforts.
- Worked with high school teachers at Loudoun Academies of Science to develop a machine learning course for high school students, 2018. Volunteer to provide project mentorship to students in these classes, 2018-present.
- Participated in "Raising Excitement for Science, Engineering, and Technology" (RESET) Education Outreach Program, 2018-2020, program organized by lab member Alice Robie.
- Taught/mentored in several summer and winter schools, including Neuromatch, FENS, Cajal, Cold Spring Harbor, Jackson Laboratory, and FLiACT. Example Colab notebooks developed for this purpose: `https://bit.ly/jaxpose`, `https://github.com/kristinbranson/roshambo_transformer`.

Publications

1. K. A. Horstmann, E. Lin, A. A. Robie, J. J. Sun, **K. Branson**. "A case study of evaluating AI agents on a neuroscience data-to-discovery pipeline." *arXiv* 2026. arXiv:2606.07718
2. L.-Q. Zhang, **K. Branson**. "Neurodata Without Boredom: Benchmarking Agentic AI for Data Reuse." *arXiv* 2026. arXiv:2605.12808
3. E. Eyjólfsdóttir, **K. Branson**. "Agent-Centric Animal Pose Forecasting." *In review*, 2026.
4. A. Sarkar, E. Lin, Z. Liu, **K. Branson**, J. J. Sun. "AgentSSL: Can MLE Agents Leverage Unlabeled Data?" *In review*, 2026.
5. G. M. Rubin, C. Managan, M. Dreher, E. Kim, S. Miller, K. Boone, A. Robie, A. L. Taylor, **K. Branson**, C. E. Schretter, A. G. Otopalik. "Networks of sexually dimorphic neurons that regulate social behaviors in *Drosophila*." *bioRxiv* 2025. doi:10.1101/2025.10.21.683766
6. H. Chiu, A. A. Robie, **K. Branson**, T. Vippa, S. Epstein, G. M. Rubin, D. J. Anderson, C. E. Schretter. "Cell type-specific contributions to a persistent aggressive internal state in female *Drosophila*." *eLife*, 2025. doi:10.7554/eLife.88598.3
7. J. A. Keller, I. S. Kwak, A. K. Stark, M. Pachitariu, **K. Branson**, J. T. Dudman. "Cortical control of innate behavior from subcortical demonstration." *bioRxiv*, 2025. doi:10.1101/2025.02.12.637930
8. R. Vaxenburg, I. Siwanowicz, J. Merel, A. A. Robie, C. Morrow, G. Novati, Z. Stefanidi, G. M. Card, M. B. Reiser, M. M. Botvinick, **K. Branson**, Y. Tassa, S. C. Turaga. "Whole-body simulation of realistic fruit fly locomotion with deep reinforcement learning." *Nature*, vol. 643, pp. 1312–1320, 2025. doi:10.1038/s41586-025-09029-4
9. A. A. Robie, A. L. Taylor, C. E. Schretter, M. Kabra, **K. Branson**. "The Fly Disco: Hardware and software for optogenetics and fine-grained fly behavior analysis." *bioRxiv*, 2024. doi:10.1101/2024.11.04.621948
10. C. E. Schretter, T. H. Sten, N. Klapoetke, M. Shao, A. Nern, M. Dreher, D. Bushey, A. A. Robie, A. L. Taylor, **K. Branson**, A. Otopalik, V. Ruta, G. M. Rubin. "Social state alters vision using three circuit mechanisms in *Drosophila*." *Nature*, vol. 637, pp. 646–653, 2024. doi:10.1038/s41586-024-08255-6
11. B. Gorko, I. Siwanowicz, K. Close, C. Christoforou, K. L. Hibbard, M. Kabra, ... **K. Branson**, G. Ihrke, S. J. Huston. "Motor neurons generate pose-targeted movements via proprioceptive sculpting." *Nature*, vol. 628, no. 8008, pp. 596–603, 2024. doi:10.1038/s41586-024-07222-5
12. M. Isaacson, J. Eliason, A. Nern, E. Rogers, G. Lott, T. Tabachnik, W. Rowell, A. Edwards, W. Korff, G. Rubin, **K. Branson**, M. B. Reiser. "Small-field visual projection neurons detect translational optic flow and support walking control." *bioRxiv* 2023.06.21.546024, 2023. doi:10.1101/2023.06.21.546024
13. J. J. Sun, A. Ulmer, D. Chakraborty, B. Geuther, E. Hayes, H. Jia, V. Kumar, Z. Partridge, A. Robie, C. E. Schretter, C. Sun, K. Sheppard, P. Uttarwar, P. Perona, Y. Yue, **K. Branson**, A. Kennedy. "The MABe22 Benchmarks for Representation Learning of Multi-Agent Behavior." *ICML*, 2023. doi:10.48550/arXiv.2207.10553
14. J. Z. Guo, B. Sauerbrei, J. Cohen, M. Mischiatti, A. Graves, **K. Branson**, A. Hantman. "Disrupting cortico-cerebellar communication impairs dexterity." *eLife*, vol. 10, p. e65906, 2021. doi:10.7554/eLife.65906
15. C. E. Schretter, Y. Aso, M. Dreher, A. A. Robie, M.-J. Dolan, N. Chen, M. Ito, T. Yang, R. Parekh, **K. Branson**, G. M. Rubin. "Cell types and neuronal circuitry underlying female aggression in *Drosophila*." *eLife*, vol. 9, p. e58942, 2020. doi:10.7554/eLife.58942
16. D. J. Im, I. S. Kwak, **K. Branson**. "Evaluation metrics for behavior modeling." *arXiv* 2007.12298, 2020. doi:10.48550/arXiv.2007.12298
17. I. S. Kwak, D. Kriegman, **K. Branson**. "Detecting the starting frame of actions in video." *WACV*, 2020. doi:10.1109/WACV45572.2020.9093405
18. B. Sauerbrei, J.-Z. Guo, M. Mischiati, W. Guo, M. Kabra, N. Verma, B. Mensch, **K. Branson**, A. Hantman. "Cortical pattern generation during dexterous movements is input-driven." *Nature*, vol. 577, pp. 386–391, 2019. doi:10.1038/s41586-019-1869-9
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20. J. M. Ache, S. Namiki, A. Lee, **K. Branson**, G. M. Card. "State-dependent decoupling of sensory and motor circuits underlies behavioral flexibility in *Drosophila*" *Nature Neuroscience*, vol. 22, no. 7, pp. 1123–1131, s41593-019-0413-4, 2019. doi:10.1038/s41593-019-0413-4

21. D. J. Im, N. Verma, **K. Branson**. "Stochastic neighbor embedding under f-divergences." *arXiv* 1811.01247, 2018. doi:10.48550/arXiv.1811.01247
22. D. J. Im, H. Ma, G. W. Taylor, **K. Branson**. "Quantitatively evaluating GANs with divergences proposed for training." *ICLR*, 2018. 10.48550/arXiv.1803.01045
23. K. McDole, L. Guignard, F. Amat, A. Berger, G. Malandain, L. A. Royer, S. C. Turaga, **K. Branson**, P. J. Keller. "In toto imaging and reconstruction of post-implantation mouse development at the single-cell level." *Cell*, vol. 175, no. 3, pp. 859–876, 2018. doi:10.1016/j.cell.2018.09.031
24. I. F. Rodriguez, R. Megret, R. Egnor, **K. Branson**, J. L. Agosto, T. Giray, E. Acuna. "Multiple animals tracking in video using part affinity fields." *Visual Observation and Analysis of Vertebrate And Insect Behavior*, 2018. [link]
25. I. F. Rodriguez, **K. Branson**, E. Acuna, J. L. Agosto-Rivera, T. Giray, R. Megret. "Honeybee detection and pose estimation using convolutional neural networks." *Reconnaissance des Formes, Image, Apprentissage et Perception*, 2018. [link]
26. **K. Branson**. "A deep (learning) dive into a cell." *Nature Methods*, vol. 15, no. 4, p. 253, 2018. doi:10.1038/nmeth.4658
27. A. A. Robie, J. Hirokawa, A. W. Edwards, L. A. Umayam, A. Lee, M. L. Phillips, G. M. Card, W. Korff, G. M. Rubin, J. H. Simpson, M. B. Reiser, **K. Branson**. "Mapping the neural substrates of behavior." *Cell*, vol. 170, no. 2, pp. 393–406, 2017. doi:10.1016/j.cell.2017.06.032
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32. S. R. Egnor, **K. Branson**. "Computational analysis of behavior." *Annual Review of Neuroscience*, vol. 39, pp. 217–236, 2016. doi:10.1146/annurev-neuro-070815-013845
33. J.-Z. Guo, A. R. Graves, W. W. Guo, J. Zheng, A. Lee, J. Rodriguez-Gonzalez, N. Li, J. J. Macklin, J. W. Phillips, B. D. Mensh, **K. Branson**, A. Hantman. "Cortex commands the performance of skilled movement." *eLife*, vol. 4, p. e10774, 2015. doi:10.7554/eLife.10774
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37. N. Verma, **K. Branson**. "Sample complexity of learning mahalanobis distance metrics." *NIPS*, pp. 2584–2592, 2015. doi:10.48550/arXiv.1505.02729
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39. **K. Branson**. "Distinguishing seemingly indistinguishable animals with computer vision." *Nature Methods*, vol. 11, no. 7, pp. 721–723, 2014. doi:10.1038/nmeth.3004
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based action selection in *Drosophila*." *eLife*, vol. 3, p. e04580, 2014. doi:10.7554/eLife.04580
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49. **K. Branson**, S. Belongie. "Tracking multiple mouse contours (without too many samples)." *CVPR*, vol. 1, pp. 1039–1046, 2005.doi:10.1109/CVPR.2005.349
50. **K. Branson**, V. Rabaud, S. Belongie. "Three brown mice: See how they run." *VS-PETS Workshop at ICCV*, 2003. [link]
51. G. W. Cottrell, K. M. Branson, A. J. Calder. "Do expression and identity need separate representations?" *Proceedings of the Annual Meeting of the Cognitive Science Society*, vol. 24, no. 24, 2002. [link]

Mentorship

Name	*Years in lab*	*Degree received*	*Current Position*
Alice Robie	2010-present		Senior Scientist in my lab
Mayank Kabra	2011-2013		Machine learning consultant, Kabra Consulting
Marta Rivera-Alba	2013-2016		Head of Data Science at Causal Foundry
Nakul Verma	2013-2017		Professor, Computer Science, Columbia U.
Kelly Seagraves	2015-2016		Senior Advisor in the U.S. Department of State's Office of the Special Envoy for Critical and Emerging Technology
Roian Egnor	2015-2026		Head of IACUC, HHMI Janelia
Iljung (Sam) Kwak	2015-2023	PhD, CS, UCSD, 2019	3D Machine Learning Engineer at Nuwa
Jiwoong (Daniel) Im	2016-2020		PhD student, CS, NYU
Rutuja Patil	2017-2024		Software Engineer in Edge Computing
Ivan (Felipé) Rodriguez	2017-2018	MS, CS, U. Puerto Rico, 2019	PhD student, Cognitive Science, Brown
Lingqi Zhang	2023-present		Theory Fellow with my lab
Aniket Ravan	2024-present		Machine Learning Researcher in my lab
Eyrun Eyjolfsdottir	2024-present		Machine learning consultant
Kai Horstmann	2025-present		PhD student, CS, Cornell
Yijia Dai	2025-present		PhD student, CS, Cornell
Manan Lalit	2026-present		Machine Learning Researcher 50% in my lab

With members of my lab, I've also co-mentored two undergraduate students and three high school students. 2/3 of my mentees have been from groups historically marginalized in STEM. As Head of Computation and Theory, I

provided mentorship and advice to the other 9 Group Leaders within this Research Area.

Outside the lab

- I love rock climbing, hiking, and generally being outside in the sun. My favorite is brainstorming zany machine learning and science ideas while doing these activities with other scientists.
- I enjoy word puzzles, particularly crossword puzzles.